

Audio Diary Engagement and Themes Across Psychiatric Diagnoses and Outcomes: Preliminary Data from the PREDiCTOR Study



Maya Valenzano¹, Joseph T. Colonel ¹, Theodore Servedio¹, Bailey Todtfeld¹, Nazly Suarez ¹, Sam Edwards¹, Adam Davidson¹, Heather Thibeau¹, Rachel Jespersen¹, Yulia Landa¹, Justin T. Baker⁴, Yoonho Chung⁴, David Kimhy¹, Kely Norel³, Habib Rahimi⁴, Sean Ryan⁵, John Torous⁵, Guillermo Cecchi³, Cheryl M. Corcoran¹, René S. Kahn¹, Shalaila S. Haas¹

¹ Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY, USA
² Department of Artificial Intelligence and Human Health, Hasso Plattner Institute for Digital Health, Icahn School of Medicine at Mount Sinai, New York, NY, USA
³ IBM Watson Research Center, Yorktown Heights, NY, USA
⁴ McLean Hospital, Harvard University, Belmont, Massachusetts
⁵ Division of Digital Psychiatry, Department of Psychiatry, Beth Israel Deaconess Medical Center, Boston, MA, USA



BACKGROUND

Digital diary tools provide a valuable opportunity to collect data on patient experiences and emotions in real time to predict diagnoses and outcomes in treatment.^{1, 2, 3} Existing research primarily examines diaries as therapeutic interventions rather than predictive tools and focuses on written diaries with structured questions.^{4, 5} Little research exists on the predictive utility of audio diaries with open-ended prompts which allow patients to elaborate on topics of their choice.⁶ The PREDiCTOR study collects audio diary speech samples from a transdiagnostic patient population ages 15-45 beginning outpatient psychiatric care, to build an AI model that predicts adverse outcomes (ED visits, hospitalization, and treatment disengagement). This project leverages data collected from the PREDiCTOR study to assess audio diary engagement as a predictive tool to identify prominent characteristics within psychiatric diagnoses and help personalize individual care.

METHODS

Demographics

Table 1. Demographic and clinical characteristics of the whole sample.			
	N = 103		N = 103
Age, mean (SD)	28.8 (7.76)	Ethnicity	
Sex		Hispanic or Latino N (%)	37 (35.9%)
Male N (%)	33 (32.0%)	Non-Hispanic or Latino N (%)	65 (63.1%)
Female N (%)	70 (68.0%)	Other or Unknown N (%)	1 (1.0%)
Race		Diagnoses	
American Indian / Alaska Native N (%)	1 (1.0%)	Anxiety Disorders N (%)	25 (24.3%)
Asian N (%)	17 (16.5%)	Depressive and Bipolar Disorders N (%)	51 (49.5%)
Black or African American N (%)	23 (22.3%)	Schizophrenia Spectrum and Other Psychotic Disorders N (%)	6 (5.8%)
White N (%)	34 (33.0%)	Trauma- and Stressor- Related Disorders N (%)	17 (16.5%)
More than one race N (%)	16 (15.5%)	ADHD and Externalizing Disorders N (%)	4 (3.9%)
Native Hawaiian or Other Pacific Islander N (%)	1 (1.0%)		
Other N (%)	11 (10.7%)		

Inclusion & Exclusion Criteria

- PREDiCTOR study participants that have completed at least 1 audio diary on their own were included (N = 103).
- Participants with Substance Use Disorders, OCD, Personality Disorders, and no diagnosis were excluded due to insufficient sample size (N = 9).

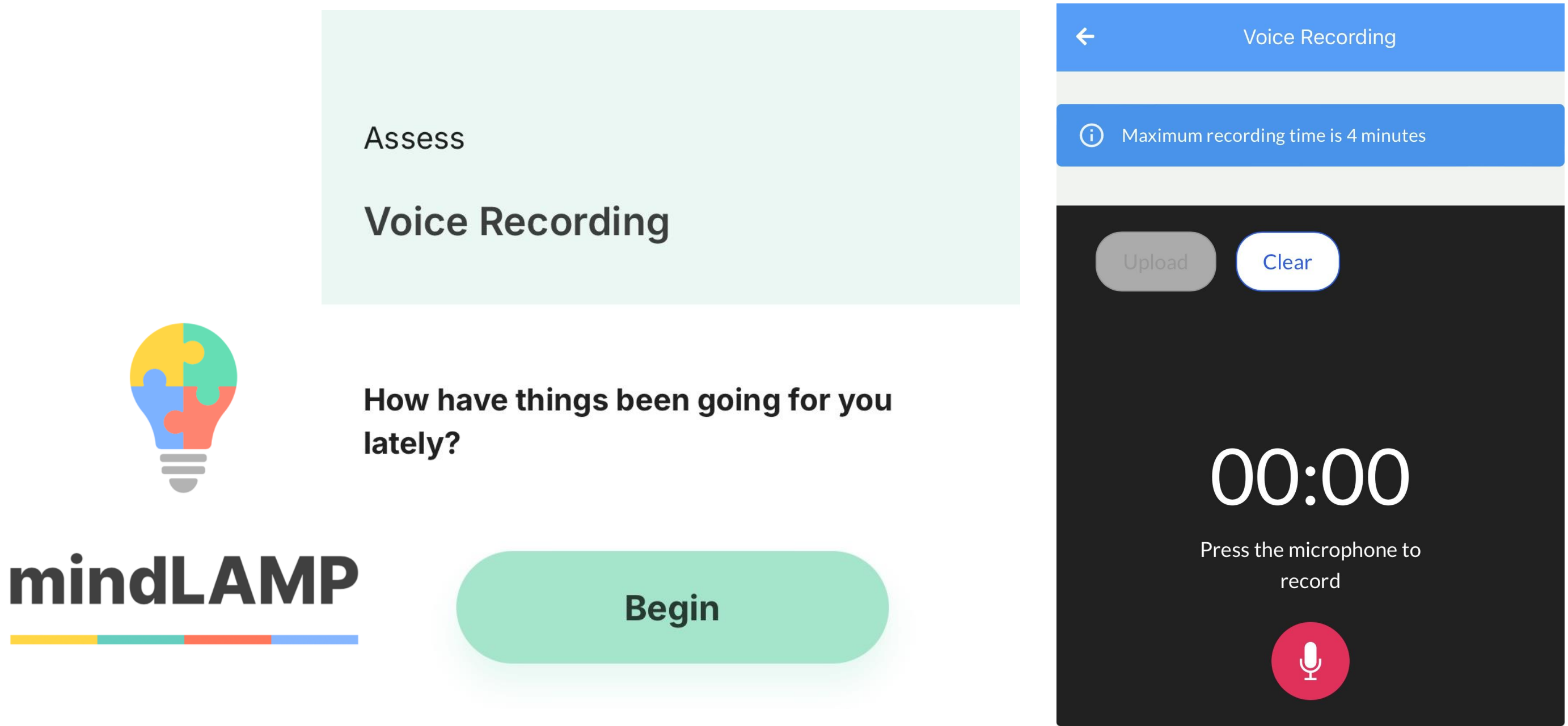


Figure 1. Audio diary prompts in the MindLAMP 2 app interface.

RESULTS

Engagement

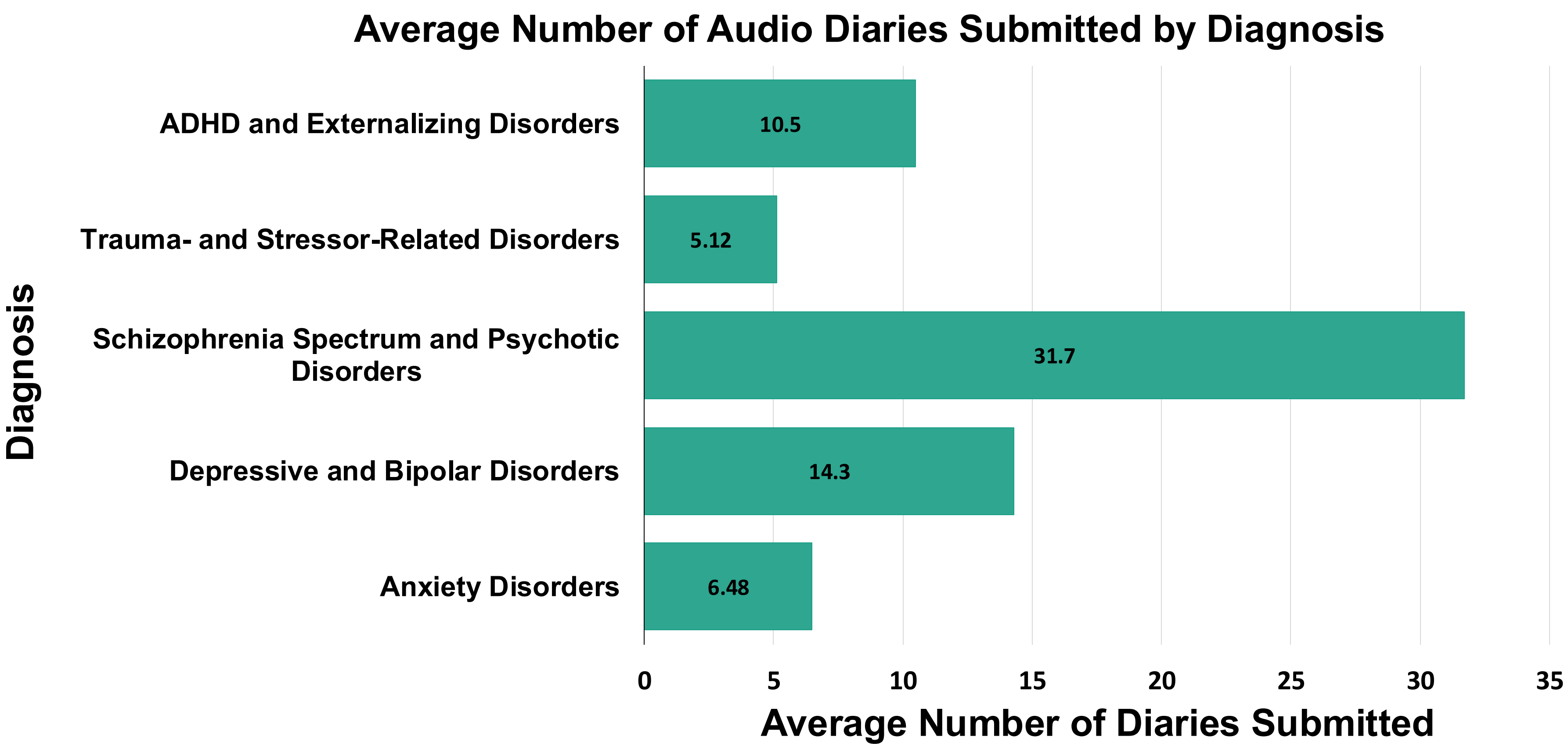


Figure 2. Mean number of audio diaries submitted by diagnosis group, controlling for number of days with the app installed.

An ANCOVA examined diagnostic group differences in the number of audio diary submissions, while controlling for number of days with the app installed. There are significant group differences in the number of audio diary submissions ($F = 3.42$, $p = 0.01$), with Schizophrenia Spectrum and Psychotic Disorders showing the highest average submissions, (mean[SD] = 31.7[40.5]), followed by Depressive and Bipolar Disorders (mean[SD] = 14.3[21.0]), ADHD and Externalizing Disorders (mean[SD] = 10.5[9.11]), Trauma- and Stressor- Related Disorders (mean[SD] = 5.12[5.94]), then Anxiety Disorders (mean[SD] = 6.48[11.5]).

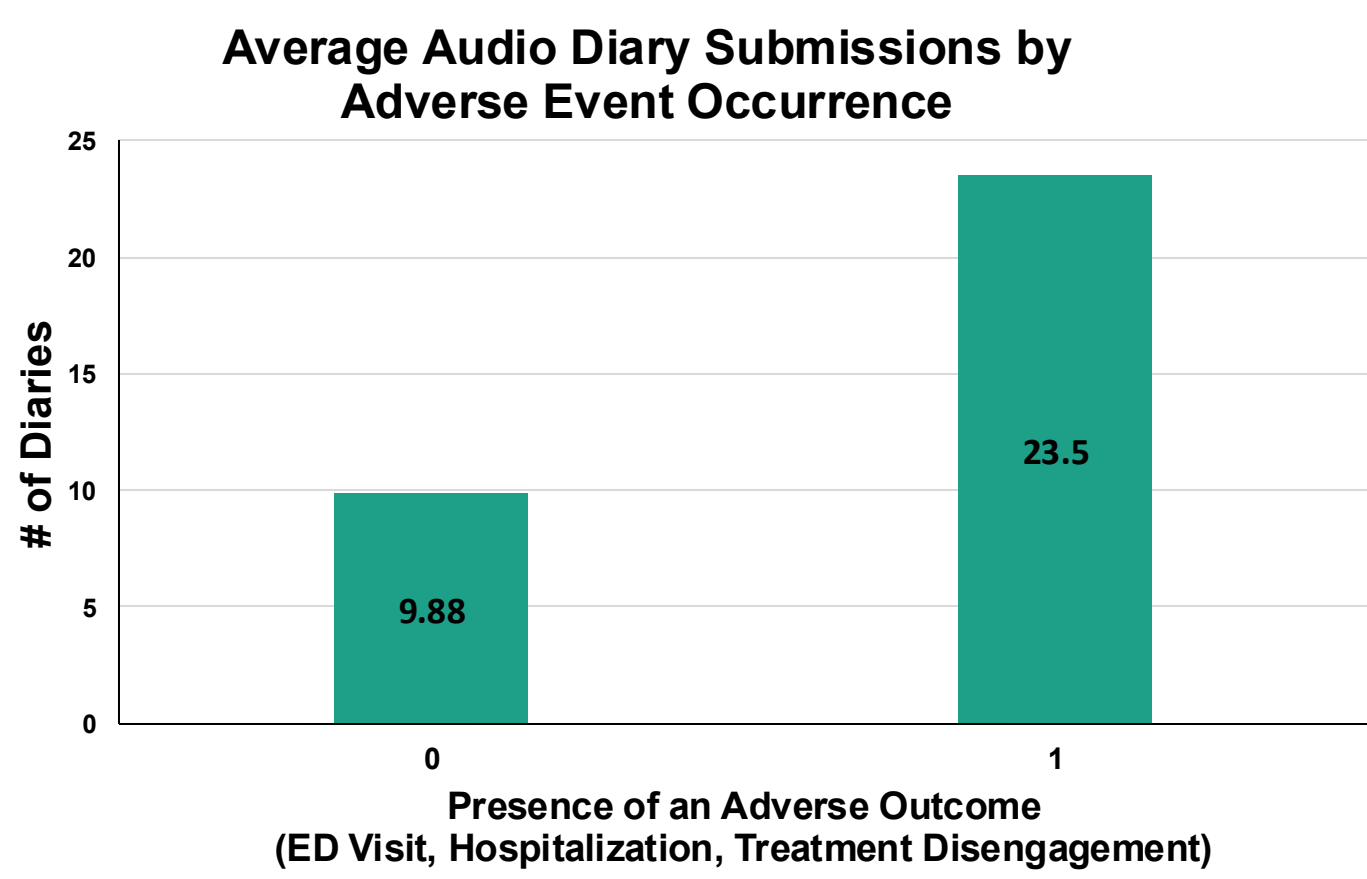


Figure 3. Mean number of audio diaries submitted by adverse event ($F = 4.77$, $p = .03$)

CONCLUSIONS

Preliminary findings suggest that psychiatric diagnoses and adverse outcomes affect engagement with audio diaries submissions. At the diagnostic level, participants with Schizophrenia Spectrum and Psychotic disorders submitted the highest number of audio diaries, and across diagnoses, both common and unique themes emerged. Additionally, participants who experienced an adverse event since enrolling in the PREDiCTOR study submitted significantly more audio diaries, which may serve as a coping mechanism prior to the onset of an adverse event. These findings have important implications for improving crisis detection, as a rapid increase in audio diary submissions could potentially be used to alert clinicians when time-sensitive intervention may be needed.

Themes

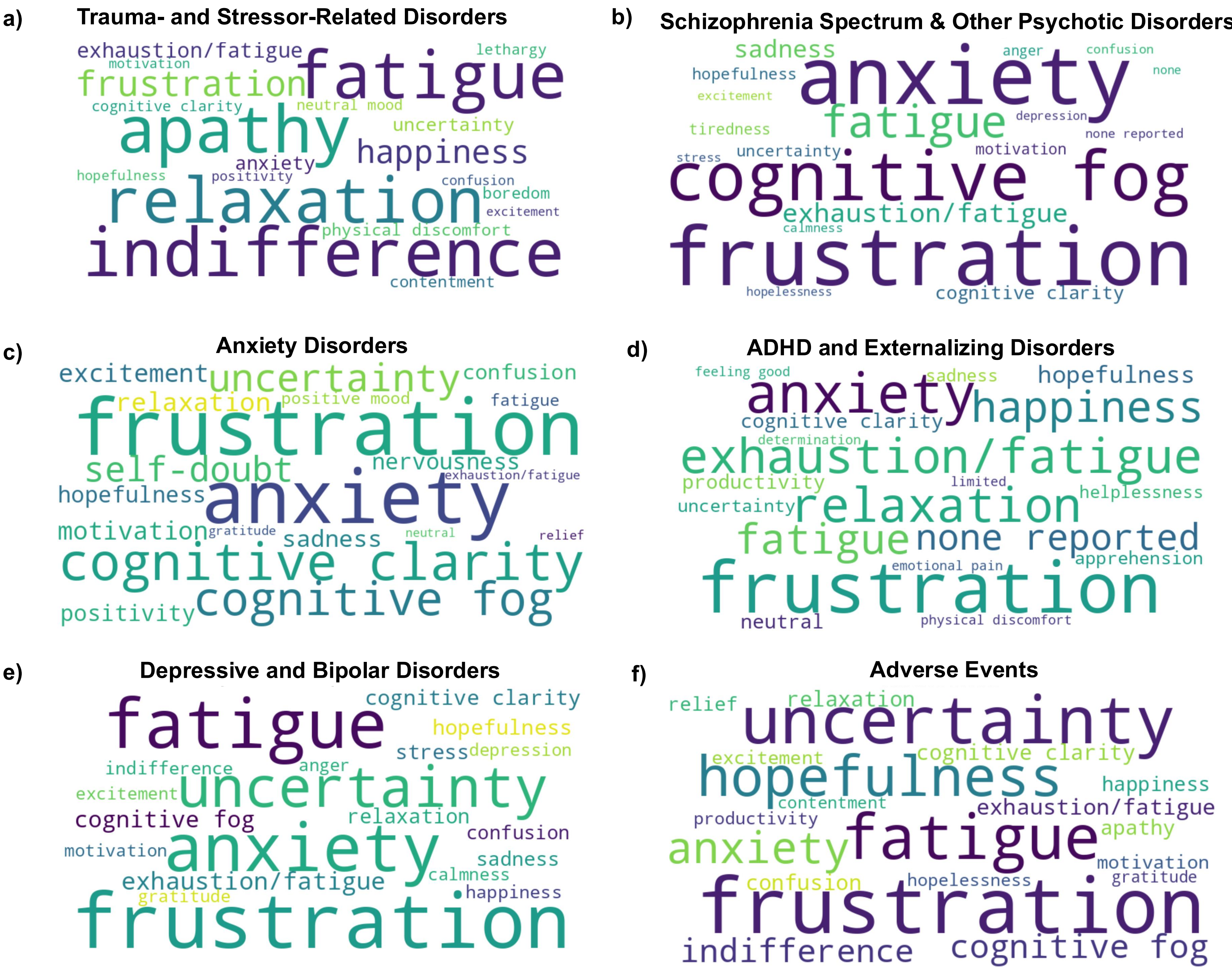


Figure 4 (a-f). Theme clouds summarizing audio diary content for each diagnosis, and whether an adverse event occurred since study enrollment.

REFERENCES AND ACKNOWLEDGMENTS

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